# From Speakers to Dialects: An Acoustic Study of Northern and Southern Peninsular Spanish Using Common Voice

**Cristina Gutiérrez Alcalá**
Speech Signal Processing
2026

## Abstract

This study compares Northern and Southern Peninsular Spanish acoustically using dialectally labeled recordings from Mozilla Common Voice, with forced alignment for phonemic measurement and statistical inference at the speaker level. Across 957 speakers and 89,958 vowel tokens, no difference was detected between the two varieties in the vowel space: no value exceeded $|d| = 0.17$ in a design powered for $d \geq 0.24$. The fact that the result is attributable to the dialects and not to the method is corroborated by an external Mexican group, which does show a clear separation from both Peninsular varieties with effects 2.5 times greater. The variation is localized in the consonant: Southern speech weakens /s/ in coda ($d = +0.63$), with /s/ at the onset serving as the null control. The study also documents how an earlier version of this same analysis yielded effects exceeding one standard deviation that turned out to stem from two individual speakers, and what is required of a volunteer corpus to avoid reproducing that error.

## 1 Introduction

The fact that a speaker from Bilbao and one from Seville sound different is an intuition shared by all Spanish speakers and one that dialectology has described for over a century. Translating that intuition into a reproducible acoustic measure is a problem that remains largely unresolved, and it is the one addressed in this paper.

The first two formants of a vowel, F1 and F2, are the resonance frequencies of the vocal tract that directly encode its height and frontness, and these constitute the standard representation of a variety's vowel space. The initial question was whether the vowel space of Northern peninsular Spanish differs measurably from that of Southern peninsular Spanish, and whether that difference is sufficient to classify an unknown speaker. Mozilla Common Voice provided an unusual resource for answering these questions: thousands of recordings with dialect labels declared by the speakers themselves, openly available and on a scale that laboratory phonetics rarely achieves.

An initial version of the analysis answered "yes", and emphatically so: differences exceeding one and a half standard deviations in F1 and in the bandwidth of F2, with $p$-values below 0.001. A subsequent audit showed that those effects stemmed from two specific speakers, who together accounted for 54% and 80% of their respective dialect cells. The study did not compare two varieties; it compared two people.

This report presents an analysis reconstructed from that finding incorporating it as the subject of study rather than as an error to be concealed. The study establishes four findings. First, it documents the mechanism by which clip-based sampling generated large, spurious effects, and shows that the magnitude obtained is a direct function of the extent to which an individual is allowed to dominate the sample (Section 5). Second, using a balanced design with 957 speakers, it finds no difference whatsoever between the two peninsular varieties in the vowel space, in a contrast with sufficient power to rule out anything other than a small effect (Section 6.1). Third, it verifies that this absence is not due to the coarseness of the method by incorporating Mexican Spanish as an outgroup, which does separate from both varieties using the same procedure (Section 6.2). And fourth, it finds a substantial effect in the consonant system: the weakening of /s/ in coda in Southern speech (Section 6.3).

The common thread linking the four is a matter of design rather than statistics, and is stated in Section 7: in a corpus built from unequal contributions by volunteers, the contrasts identified within individual speakers are robust, while the overall comparisons between groups are fragile.

## 2 State of the art

### 2.1 Dialectal variations in Peninsular Spanish

Spanish has a remarkably stable system of five vowels: /a, e, i, o, u/. It lacks distinctions based on vowel length and tension; its unstressed vowels are not reduced as they are in English or Portuguese; and its inventory has hardly changed since the Romance language system became fixed. This stability has a direct implication for this study: the vowel system offers little scope for dialectal differentiation, and variation in Spanish is overwhelmingly consonantal (Hualde, 2005; Martínez Celdrán and Fernández Planas, 2007). The reference formant values for peninsular Spanish were established by Quilis (1981) based on laboratory speech and remain the standard basis for comparison.

The best-documented feature of Southern speech is the weakening of /s/ in implosive position, which is realized as aspiration, assimilation to the following segment, or elision, in contrast to the characteristic retention found in the peninsular North (Alvar, 1996; Narbona Jiménez et al., 1998; Gerfen, 2002; Ruch and Harrington, 2014). Associated with this weakening, a classic descriptive account holds that the preceding vowel compensates for the loss by opening and lengthening, to the point of positing an expanded vowel system based on degree of openness for Eastern Andalusian (Alonso et al., 1950). This assertion, based on auditory observation and small samples, can be tested directly with current methods, and Section 6.4 does so.

### 2.2 Formant Normalization

Formant frequencies depend on the length of the vocal tract as much as on articulatory configuration, so that two speakers producing the same vowel will systematically differ in hertz if they differ in anatomy. Comparing varieties in raw hertz values, when the data sets are not balanced by sex, is to confuse dialect with physiology. Sociophonetics solves this problem by normalizing data per speaker, and among the available procedures, the intra-speaker z-score proposed by Lobanov (1971) performs best in systematic comparisons: it eliminates anatomical variation while preserving sociolinguistic differences (Adank et al., 2004). The Bark scale (Traunmüller, 1990), which is perceptually based, is used here as a complementary measure.

### 2.3 Common Voice as a phonetic corpus

Common Voice is a multilingual corpus of read speech built from contributions by volunteers and distributed openly (Ardila et al., 2020). For dialectology, it offers something rare: variety labels declared by the speakers themselves, on a scale of thousands of speakers. It also presents two challenges. The first is that these labels are neither verifiable nor granular. The second, more significant challenge is that the distribution of contributions is extraordinarily uneven, because nothing in the project's design encourages equitable distributions: a few enthusiastic volunteers can each contribute hundreds of recordings. Sampling that selects clips without grouping them by speaker therefore produces observations that are not independent, with the consequent risk of pseudoreplication (Hurlbert, 1984; Winter, 2019). The availability of high-quality forced aligners for Spanish (McAuliffe et al., 2017) is what allows us to move from clip-level measures to phoneme-level measures, and thereby to examine the intra-speaker contrasts that the corpus does indeed robustly support.

## 3 Corpus and sampling

### 3.1 Common Voice 26.0 and dialect tags

All the data used come from Mozilla Common Voice, Spanish 26.0 version (12 of June of 2026), a corpus of read speech built from voluntary submissions and distributed under an open licence (Ardila et al., 2020). The version used contains 439,225 validated clips. Each clip has an orthographic transcription associated with it and a speaker identification number (`client_id`), in addition to autodeclared metadata such as gender, age and accent.

Getting the corpus onto a machine where I could work with it shaped the pipeline more than I expected. The Spanish release is a single archive of roughly 48 GB. My first attempt was to download it, unpack it and work from the audio. That cost two days of failed transfers and still left me without room for the extracted files. The fix turned into the design. Nothing in the sampling stage needs the audio, because the selection is made entirely from `validated.tsv`, which sits at the front of the archive and can be pulled out with `tar --fast-read` in about a second. Only the clips the sampler selects are then extracted, which in the final study meant unpacking 5,577 files instead of 439,225.

The accent tag is what makes this corpus usable in a dialectal study, since the speakers are responsible for tagging the dialectal variation that they consider they speak. From the available Spanish tags, I selected three: "peninsular septentrional" also referred to as North, "peninsular meridional" or South and Mexican. The latter being incorporated as an external group due to the reasons explained in Section 6.2. The speakers are also able to mark one or more accents, and a non-negligible proportion do. I decided to discard those clips instead of manually assigning one of the tags, since any tie-breaker criterion would introduce a biased opinion in the grouping variable. After the filter was applied, I had 236,832 clips with exactly one of the aforementioned tags.

The nature of these tags must also be mentioned. They are autodeclarations and not dialectological classifications. There is no independent verification of the provenance of the speaker, or further fine-grained information than the region. This limitation will be touched on in Section 8.

### 3.2 Speaker-level sampling and per-speaker cap

The most important design choice of this project was also the simplest: the sampling was done directly on the speaker and not on the clip. An older version of this analysis selected clips randomly, identifying them by file path, without looking at who had produced each clip. The result was a sample dominated by a handful of individuals. Section 5 is devoted entirely to the consequences.

The current sampling groups first by `client_id`, arranges the clips of each speaker by a reproducible criterion and applies a cap of eight clips per speaker. The declared design objective is to first maximize the number of speakers and then the number of clips per speaker, because that same number of speakers (and not number of tokens) is what narrows down inference. The whole sampling stage is seeded, and is registered in a versioned manifest, so that the sample is reconstructible clip by clip.

The cap of eight is a compromise between two requirements that pull in opposite directions. Lobanov normalization needs enough vowels from each speaker to estimate that speaker's own mean and spread, which argues for more clips per person. Independence argues for fewer. Eight clips of roughly five seconds yield on the order of a hundred vowel tokens per speaker, comfortably enough for the normalization, while holding any single individual's share of a dialect cell below one percent. The sampling is implemented with pandas (McKinney, 2010) over NumPy arrays (Harris et al., 2020).

| Variety | Clips | Speakers | With tokens | Normalizable | Tokens |
|---|---|---|---|---|---|
| North | 2,639 | 469 | 468 | 443 | 42,805 |
| South | 1,199 | 188 | 188 | 181 | 19,549 |
| Mexico | 1,739 | 300 | 299 | 282 | 27,604 |
| Total | 5,577 | 957 | 955 | 906 | 89,958 |

Table 1: Composition of the sample in each variety.

One consequence of prioritizing speakers is that the design is bounded by its smallest group. Southern Peninsular Spanish has 188 speakers in release 26.0, and that number constrains everything the study can claim, so Northern and Southern speakers were both taken in full. Mexican Spanish has over 1,500 available, far more than the comparison needs, and was capped at 300.

The effect of the cap is measured with the Gini coefficient of the distribution of clips per speaker, where 0 indicates a perfectly even distribution and 1 a total concentration. In the preliminary sample the Gini coefficient was 0.76 in North and 0.85 in South. In the current sample it is 0.26 in North and 0.18 in South. This sole change is what made the rest of the analysis interpretable.

### 3.3 Final design of the sample

The resulting sample has 5,577 clips of 957 different speakers, and a total of 484.5 recorded minutes. Table 1 compiles its composition.

The imbalance between varieties is dictated by the corpus itself. The Southern part of the peninsula is represented by fewer than half as many speakers as the North. I do not correct for this by undersampling the North, because discarding valid data solely to balance the cells would reduce statistical power without gaining anything. The comparisons are calculated using tests that do not assume equal sample sizes.

The imbalance by sex, however, does matter. All three varieties are dominated by male voices: 27.9% of female speakers in the North, 20.4% in the South, and 25.2% in Mexico, with the rest being male or undeclared. Since the length of the vocal tract directly influences formant frequencies, a comparison between these groups in raw hertz does not distinguish dialect from anatomy. This is why the normalization described in Section 4.3 is not an optional refinement but a design require-

ment.

Figure 1 summarizes the complete path, from the 439,225 validated clips in the corpus to the tokens finally analyzed. The alignment and quality-control stages that produce those numbers are described in Sections 4.1 and 4.3.

The sample of 903 clips on which the preliminary analysis was based remains unchanged and is reused in Section 5, where it serves not as a data source but as an object of study.

## 4 Methodology

### 4.1 Audio preprocessing and forced alignment

Each clip was converted from MP3 to mono WAV at 16 kHz, which is the sample frequency expected by the aligner, with peak scaling to avoid clipping. Next to each audio file, a text transcription file was written from what the speaker read aloud, taken directly from the corpus metadata, and the ensemble was organized in per speaker directories so that the aligner could apply its speaker adaptation.

Forced alignment was not part of the original plan. It became necessary after two attempts to recover vowel identity by cheaper means, and both failures explain why the pipeline looks the way it does. The first classified each measurement by its position in the F1/F2 plane, using the reference ranges for Spanish vowels in Quilis (1981). Nearly every token it labelled /a/ came from the Northern group and nearly every /i/ from the Southern one. Those ranges are calibrated on Northern speech, so a displaced Southern token lands in a neighbouring category. The procedure is also circular: labelling vowels by their formants and then comparing formants across dialects builds the conclusion into the labels. The second attempt inferred the vowel from the transcription, which works only for clips containing a single vowel type. Of 903 clips, 14 resolved.

That left aligning the audio to its transcription properly. I installed the Montreal Forced Aligner (McAuliffe et al., 2017), built on Kaldi (Povey et al., 2011), together with the pretrained `spanish_mfa` acoustic model and dictionary. The aligner takes the audio and its orthographic transcription and returns a TextGrid with the time boundaries of each word and each phone. Of the 5,577 clips, 5,572 aligned correctly (99.9%); the remaining five failed because of discrepancies between the transcription and the signal.

With alignment, each clip yields all of its vowel nuclei with phonemic labels, and with them four pieces of information that were unavailable before: vowel identity, duration, stress position, and the following segment. The last of these is what made the coda /s/ analysis of Section 6.3 possible. The semivowels /j/ and /w/ are deliberately excluded, since they are not syllable nuclei and their formant targets are transitional.

### 4.2 Extraction of acoustic parameters

The acoustic measures were taken using Praat (Boersma and Weenink, 2021) through its Python interface Parselmouth (Jadoul et al., 2018), using linear predictive analysis with Burg's method with a maximum of five formants.

The frequency ceiling is adapted to the speaker's declared sex: 5,000 Hz for male voices, 5,500 Hz for female voices, and 5,250 Hz when the corpus records no value. The previous analysis applied a singular ceiling of 5,000 Hz to all of the speakers, which systematically biased the estimation of female formants. This was not a trivial problem, since the dialect cells are not balanced by sex.

From each vowel, F1, F2, B1 and B2 were registered in three different points: 25%, 50%, and 75% of its duration. The middle point is the conventional measure; the other two allow to quantify the formant movement inside a vowel, relevant to descriptions of diphthongization in Southern speech. To them, I added segment duration, fundamental frequency, the harmonics-to-noise ratio (HNR), the clip's signal-to-noise ratio, lexical stress and the following consonantal context. The vowels with a duration less than 30 ms were discarded from the start, since they were already dominated by the transitions towards the adjacent consonants and lacked a measurable stationary state. The extraction was parallelized on various processes.

### 4.3 Quality control and normalization

Quality control first applies deliberately generous range filters: F1 between 150 and 1,100 Hz, F2 between 500 and 3,200 Hz, and bandwidth between 10 and 900 Hz (B1) and 10 and 1,000 Hz (B2). These limits exist to remove tracking failures (an F1 of 3 kHz is an octave error, not a vowel), not to impose a phonetic expectation. Tightening them further would reintroduce the circularity that invalidated the vowel tagging of the preliminary analysis. Also, F2 must be higher than F1 and outliers beyond three standard deviations within each speaker and each vowel were eliminated, a filter that is only

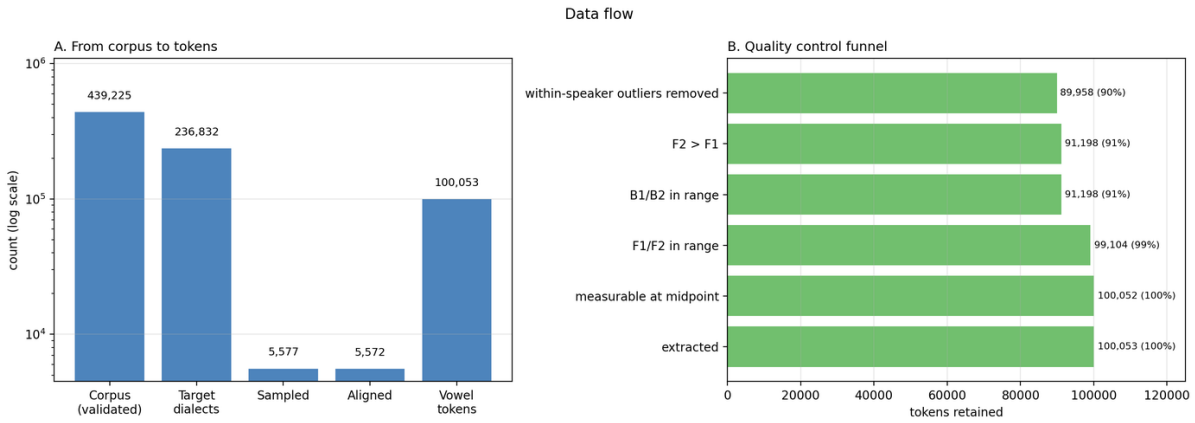



Figure 1: From corpus to analyzed tokens. The left panel shows the reduction from the validated Common Voice clips to the final sample after filtering by unique label and applying the limit of eight clips per speaker. The right panel shows the quality control funnel applied to the extracted vowel tokens.

| Phase | Tokens | % retained |
|---|---|---|
| Extracted | 100,053 | 100.0 |
| Measurable at midpoint | 100,052 | 100.0 |
| F1/F2 in range | 99,104 | 99.1 |
| B1/B2 in range | 91,198 | 91.1 |
| F2 > F1 | 91,198 | 91.1 |
| Within-speaker outliers removed | 89,958 | 89.9 |

Table 2: Quality control funnel on the extracted vowel tokens.

possible when the identity of the segment is known.

The main loss is concentrated in the bandwidth filters, which discards 8% of tokens. The bandwidths estimated by linear prediction were notably unstable and sensitive to the recording conditions, a fact that is relevant in Section 5, where one of the preliminary analysis effects was precisely an effect of B2.

On the surviving tokens, three representations were calculated. The first is raw hertz, which is retained for transparency but does not support any comparisons between groups, for the reasons explained in Section 3.3. The second is the Bark scale (Traunmüller, 1990), which approximates the ear's frequency resolution. The third, and the one used in all contrasts, is Lobanov normalization (Lobanov, 1971): the z-score of each formant computed within each speaker across all of their vowels. It is the best-performing procedure in systematic comparisons of normalization methods (Adank et al., 2004), for the reasons set out in Section 2.2. It requires that the speaker has produced all five vowel categories, a condition met by 906 of the 955 speakers with tokens. The remaining 49 are excluded from the normalized analyses.

### 4.4 Statistical scope

The principle that governs all inference is that the unit of analysis is the speaker (Hurlbert, 1984; Winter, 2019). Tokens from the same speaker are not independent observations. They all share the same anatomy, microphone, recording environment, and reading style. Treating them as independent inflates the degrees of freedom and produces significance regardless of whether or not a dialectal effect exists, as demonstrated empirically in Section 5.

In practice, this means that the data are aggregated into speaker $\times$ vowel cells and, from there, into a single measure per speaker, prior to any comparisons. Comparisons between varieties are performed using Welch's $t$-test, which does not assume equal variances or sample sizes, along with Cohen's $d$ (Cohen, 1988) and confidence intervals obtained via bootstrapping (Efron and Tibshirani, 1993) with 2,000 resamples of speakers, not tokens. Welch's test and the bootstrap are computed with SciPy (Virtanen et al., 2020), and the mixed models are fitted with statsmodels (Seabold and Perktold, 2010). Within each family of contrasts, $p$-values are corrected for multiple comparisons using Holm's procedure (Holm, 1979). Mixed models with a random intercept per speaker (Barr et al., 2013) are fitted to the means per speaker and vowel. The significance level is 0.05, and every stochastic step, like sampling, bootstrapping, $k$-means initialization, and cross-validation partitions, is seeded with a fixed seed.

A methodological caveat is worth noting, because it is an easy pitfall to fall into. Fitting a token-level mixed model to values already normal-

ized by Lobanov yields an ICC close to zero and $p$-values less than 0.001. Lobanov sets each speaker's mean to 0 and their standard deviation to 1, so that there is no remaining variance between speakers for a random intercept to account for, and the model reverts to treating the tokens as independent. A random intercept protects against speakers differing in level, not against one speaker shaping a within-speaker pattern. Aggregating into speaker $\times$ vowel cells before modeling is what solves the problem.

Finally, the design is accompanied by a power analysis computed from the realized speaker counts, which sets limits on what the study can honestly claim. The resulting thresholds are reported alongside the results in Section 6.1.

### 4.5 Validation of the extractor

To verify that the extractor measures what it claims to measure, vowels with known formants are synthesized using an F1–F5 resonator bank on a 120 Hz glottal source, and the recovered value is compared with the actual value. Measurement is accurate for open vowels (the error in F1 for /a/ is +0.2%) but degrades for closed vowels, with an error of +13.5% for /i/ and +33.7% for /u/. The cause is well known: in closed vowels, F1 approaches F0, so that individual harmonics dominate the fitted envelope, and in /u/, the F1 and F2 poles interfere with each other.

The bias is identical in both dialects, so it cannot generate a spurious difference between them, but it makes /u/ the least reliable measure in the study, and it is treated as such in the discussion. The test is part of a battery of 24 automated tests that also verify the handling of coda context, the identification of the stressed syllable, and the normalization functions.

The pipeline runs as a single seeded entry point; figures are produced with Matplotlib (Hunter, 2007), and TextGrids are parsed with praatio. A full run takes about fifteen minutes end to end.

## 5 Pseudoreplication

### 5.1 Naive analysis

The first version of this study worked on 440 tokens (244 North and 196 South), obtained from taking a measurement per clip in the maximum global intensity peak. Applying Welch $t$-test to those tokens, three out of four comparisons were significant, and two of them with effects of a magnitude that is rarely observed in dialectal phonetics. F1 had a $d = +1.34$ and B2 $d = -1.43$, both with $p < 0.001$. Literally interpreted, these numbers described two varieties whose vowels were separated by more than one and a half standard deviations.

The result was, in appearance, excellent. It was also false, and the way in which it was false is more instructive than any effect it might have found.

### 5.2 Audit per speaker

The 440 tokens do not come from 440 different speakers. When the measurements are cross-referenced with the speaker identifiers of the corpus, the sample revealed 54 Northern speakers and 26 Southern speakers, with an extraordinarily unequal distribution. A single speaker contributed 132 tokens, which was 54% of the septentrional cell, and another one made up 157, 80% of the meridional cell. The three most prolific speakers of each cell concentrated 64% and 86% of the data, respectively.

The consequences can be directly seen by eliminating speakers one by one. If the dominant meridional speaker is discarded, $d$(F1) falls from 1.34 to 0.23 and $d$(B2) from $-1.43$ to $-0.45$. If the septentrional speaker is discarded, they fall to 1.04 and $-1.04$. If both are removed (289 of the 440 tokens, leaving 151), the effects are reduced to $d = +0.01$ and $d = -0.09$, translated to almost nothing. The elimination of any other speaker leaves the numbers practically intact.

What the preliminary analysis measured, thus, was not the difference between septentrional and meridional Spanish, but the difference between two concrete people. This is the problem that Hurlbert (1984) called pseudoreplication. Pseudoreplication means treating as independent replicas observations which share the experimental unit. The $t$-test assumes independence; 440 clips from 80 speakers do not satisfy it, and when one of these makes up 80% of a cell, the effective sample size of that cell gets closer to one than to two hundred.

### 5.3 Size of the effect

The dependence can be directly quantified by recalculating the original analysis under increasing token caps per speaker. With a cap of one token per speaker (80 strictly independent observations) $d$(F1) is $+0.11$. With a cap of three, $+0.11$. With a cap of eight, $+0.30$. With a cap of twenty, $+0.54$; with a cap of fifty, $+0.90$; with a cap of one hundred, $+1.15$; and with no cap at all, $+1.34$. The

| | naive $d$ | $p$ (Holm) | per spk $d$ | $p$ | mixed $p$ | ICC |
|---|---|---|---|---|---|---|
| F1 | $+1.335$ | $<0.001$ | $+0.040$ | 1.000 | 1.000 | 0.54 |
| F2 | $-0.337$ | 0.001 | $+0.333$ | 0.835 | 1.000 | 0.17 |
| B1 | $-0.166$ | 0.100 | $+0.195$ | 1.000 | 1.000 | 0.05 |
| B2 | $-1.433$ | $<0.001$ | $-0.172$ | 1.000 | 1.000 | 0.28 |

Table 3: Naive analysis per token against a speaker level analysis, on 440 tokens.

relationship is monotonic, and it is replicated in B2 with the opposite sign.

In other words: the published effect size is a direct function of how much an individual is allowed to dominate the sample. There is no "true" value that the cap distorts; the cap is the sole determinant of the obtained value.

### 5.4 Naive against corrected

Table 3 contrasts the four original comparisons with two analyses that respect the data structure: the aggregation of one measure per speaker and a mixed model with a random intercept per speaker (Barr et al., 2013; Winter, 2019).

No effect survives. Note also that F2 not only loses its significance but also changes sign when moving to the speaker level, which rules out interpreting it as an attenuated real effect.

The ICC for F1 is 0.54: more than half of the variance in F1 is explained by who is speaking, not by which dialect they speak. From this value, the design effect can be calculated, indicating that the 440 pooled tokens carry the information of approximately 128 independent observations. The bootstrap-constructed confidence intervals show the same result from another angle. When resampling tokens, $d$(F1) has an interval of $[+1.11, +1.60]$, which comfortably excludes zero. When resampling speakers, the interval is $[-0.20, +1.83]$ and includes it. For B2, $[-1.67, -1.22]$ versus $[-1.89, +0.19]$. All four speaker-resampled intervals contain zero (Figure 2).

### 5.5 Why the balanced result helps

When applied to the sample described in Section 3.3, the same audit procedure finds nothing to audit. The most prolific southern speaker now accounts for 1% of their cell, and their removal shifts the largest per-vowel effect by 0.00, compared to the shift from 1.6 to 0.0 produced by the same operation in the previous sample. The pooled analysis and the speaker-level analysis coincide, simply because there is nothing left on which they might disagree.

| Measurement | North | South | $d$ | $p$ | $p$ (Holm) |
|---|---|---|---|---|---|
| F1 /a/ | 1.09 | 1.07 | $+0.12$ | 0.166 | 1.000 |
| F1 /e/ | $-0.31$ | $-0.28$ | $-0.17$ | 0.043 | 0.638 |
| F1 /i/ | $-1.22$ | $-1.21$ | $-0.03$ | 0.720 | 1.000 |
| F1 /o/ | $-0.09$ | $-0.08$ | $-0.04$ | 0.661 | 1.000 |
| F1 /u/ | $-0.84$ | $-0.84$ | $+0.01$ | 0.893 | 1.000 |
| F2 /a/ | $-0.25$ | $-0.27$ | $+0.14$ | 0.085 | 1.000 |
| F2 /e/ | 0.59 | 0.58 | $+0.02$ | 0.783 | 1.000 |
| F2 /i/ | 1.50 | 1.52 | $-0.06$ | 0.440 | 1.000 |
| F2 /o/ | $-0.96$ | $-0.95$ | $-0.04$ | 0.684 | 1.000 |
| F2 /u/ | $-1.10$ | $-1.16$ | $+0.11$ | 0.135 | 1.000 |
| Duration (s) | 0.077 | 0.077 | $-0.00$ | 0.995 | 1.000 |
| B1 (Hz) | 149.2 | 146.1 | $+0.05$ | 0.528 | 1.000 |
| B2 (Hz) | 218.0 | 216.8 | $+0.02$ | 0.822 | 1.000 |
| HNR (dB) | 9.97 | 9.81 | $+0.04$ | 0.616 | 1.000 |
| F0 (Hz) | 151.3 | 149.4 | $+0.05$ | 0.580 | 1.000 |

Table 4: Speaker level comparison between North and South. F1 and F2 values are normalized by Lobanov (intra-speaker z-scores); the bandwidths, the duration, the HNR and F0 are expressed using their original units. A positive value of $d$ indicates a higher value in the North.

This convergence is the argument in favor of the per-speaker cap, and it is also the reason why the results in Section 6 can be interpreted with confidence: not because the statistical method is more sophisticated, but because the sampling design no longer allows the question to answer itself.

## 6 Results

### 6.1 Vowel space on speaker level

The main comparison of this project was made on a per-speaker and per-measure average, between 456 and 461 Northern speakers and between 178 and 186 Southern speakers, according to how many produced the vowel at hand. Table 4 compiles the fifteen evaluated measures: the normalized positions of the five vowels in F1 and F2, the average duration, the bandwidths, the harmonics-to-noise ratio and the fundamental frequency.

None of these comparisons reaches significance after the correction by multiple comparisons. The major observed effect in any point on this table is $|d| = 0.17$, corresponding to an F1 of /e/, whose $p$ value without correcting (0.043) stays at 0.638 after applying Holm on the fifteen contrast family. The bandwidths, where the preliminary analysis located its second strongest effect ($d = -1.43$), now show $d = +0.02$.

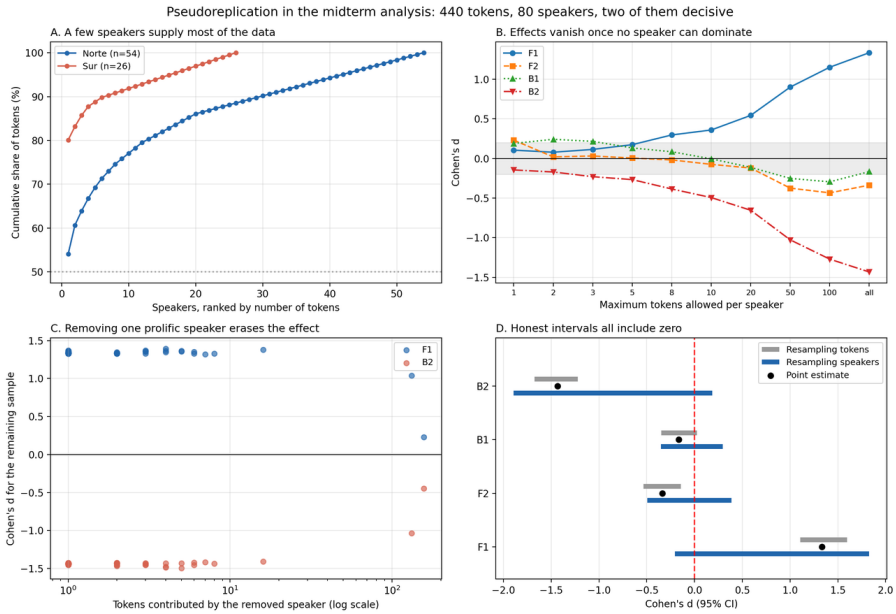



Figure 2: Anatomy of a sampling artifact, based on the 440 tokens from the preliminary analysis. A: A few speakers contribute most of the data for each dialect. B: The effects fade as the cap on tokens per speaker prevents any single individual from dominating. C: Removing a single prolific speaker eliminated the effects. D: The confidence intervals obtained by resampling speakers all include zero, unlike those obtained by resampling tokens.

The same conclusion is obtained with global vowel space measurements instead of using individual positionings (Figure 3). The area of the convex hull defined by the five normalized vowels is 3.05 in the North and of 2.95 in the South, with $d = +0.12$ and a confidence interval at 95% of $[-0.05, +0.28]$, $p = 0.167$. Mixed models fitted to the speaker $\times$ vowel means show no significant main effect of dialect.

What makes this result informative is not the absence of significance, but the power of the design. With 461 Northern speakers and 186 Southern speakers, the comparison would detect, with 80% probability, any effect of magnitude $d \geq 0.24$, and with 90% probability, any effect of $d \geq 0.28$. The sample from the preliminary analysis, by contrast, could only detect effects of $d \geq 0.63$, that is, large effects only.

The conclusion drawn from these data must be formulated with precision. It is not that the vowel spaces of both varieties are identical: differences below $d \approx 0.24$ remain entirely possible, and this study would not rule them out. What can be stated, however, is that there is no difference in the vowel space between the North and South of the Peninsula large enough to be detected by a design with more than six hundred speakers. Any effect that exists is, at most, small.

The obvious objection remains: that the method is simply incapable of distinguishing dialects. The following section addresses this objection.

### 6.2 Calibration with an external group

The result in Section 6.1 allows for two incompatible interpretations. One is that the two peninsular varieties do not differ appreciably in their vowel systems. The other is that the procedure is too crude to detect any dialectal differences, and would have produced a null result regardless of the comparison made. Without additional evidence, both explanations account for the data equally well.

The way to distinguish between them is to apply exactly the same procedure to a contrast whose existence is not in dispute. To that end, 300 Mexican speakers were included in the design and processed using the same sampling, the same alignment, the same feature extraction, the same normalization, and the same statistical tests. If the instrument distinguishes between Peninsular and Mexican Spanish but not between Northern and Southern Spanish, the absence of an effect in Section 6.1 is a statement about the dialects, not about the method.

The three pairwise contrasts, based on the twelve vowel measures evaluated at the speaker level, show the following results. North versus South gives an average $|d|$ of 0.064 and a maximum of 0.172, with 0 of the 12 comparisons significant.

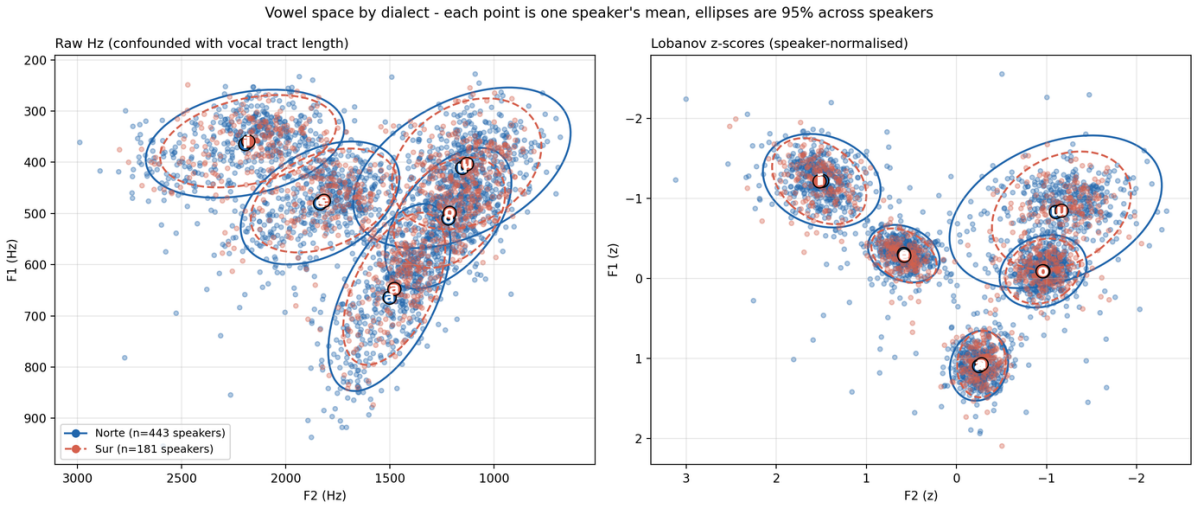



Figure 3: Vowel space of the Northern and Southern parts of the Iberian Peninsula. On the left, in raw hertz; on the right, after Lobanov normalization. Each point represents the average for a single speaker, and the ellipses cover 95% of the variation among speakers. The reduction in dispersion between the two panels corresponds to the anatomical variation that normalization eliminates; what remains is nearly a complete overlap between varieties.

North versus Mexico gives 0.152 and 0.303, with 4 of 12. South versus Mexico gives 0.173 and 0.411, with 2 of 12.

Transatlantic contrasts produce effects that are, on average, 2.5 times greater than those of the Peninsular contrast, and reach statistical significance after correction in six of the twenty-four comparisons, whereas the North-South contrast does not reach significance in any of the twelve. The largest differences are concentrated, as might be expected, in F2 of the peripheral vowels: F2 of /a/ between the South and Mexico ($d = -0.41$, $p < 0.001$), F2 of /i/ between the South and Mexico ($d = +0.40$, $p < 0.001$), and F2 of /i/ between the North and Mexico ($d = +0.30$, $p = 0.004$). Vowel duration also distinguishes the North from Mexico ($d = -0.28$, $p = 0.032$), without distinguishing the North from the South.

The conclusion is that the method does resolve dialectal differences when they exist, using the same sample, the same measures, and the same decision threshold (Figure 4). The null result in Section 6.1 is not an artifact of a blunt instrument.

It is worth saying here, rather than in the discussion, that these effects are small in absolute terms. A mean $|d|$ of 0.16 for the transatlantic contrast is not what the descriptive literature would lead you to expect from two varieties separated by an ocean and five centuries. I read that as a problem of scale rather than a failure of the comparison. The five-vowel system described in Section 2.1 gives dialectal drift very little to work with (Hualde, 2005), and the outgroup is what makes this visible: even across the Atlantic, these vowels barely move. Seen that way, the null result of Section 6.1 is close to what Spanish phonology would predict. This immediately suggests where variation should be sought instead of in the vowels. The following section explores this.

### 6.3 /s/ in coda

If dialectal variation in Spanish lies primarily in the consonants, the obvious candidate for contrast within the Iberian Peninsula is the behaviour of /s/ in syllable-final position. Descriptions of the Southern speech consistently point to its weakening (aspiration, assimilation to the following segment, or elision) in contrast to the full realization retained by Northern speech (Narbona Jiménez et al., 1998; Gerfen, 2002; Torreira, 2007).

This hypothesis has a feature that makes it particularly suitable in light of what we saw in Section 5: it is contrasted within each speaker. The relevant comparison is not between the Northern /s/ and the Southern /s/, but between the /s/ in coda and the /s/ at the onset of the same speaker. Each speaker thus acts as their own control, and an imbalance in the number of clips per speaker cannot produce the effect, because it affects both terms of the comparison equally. This is exactly the type of contrast that the preliminary analysis could not address, due to the lack of phonemic segmentation.

The alignment yields 21,639 /s/ segments from

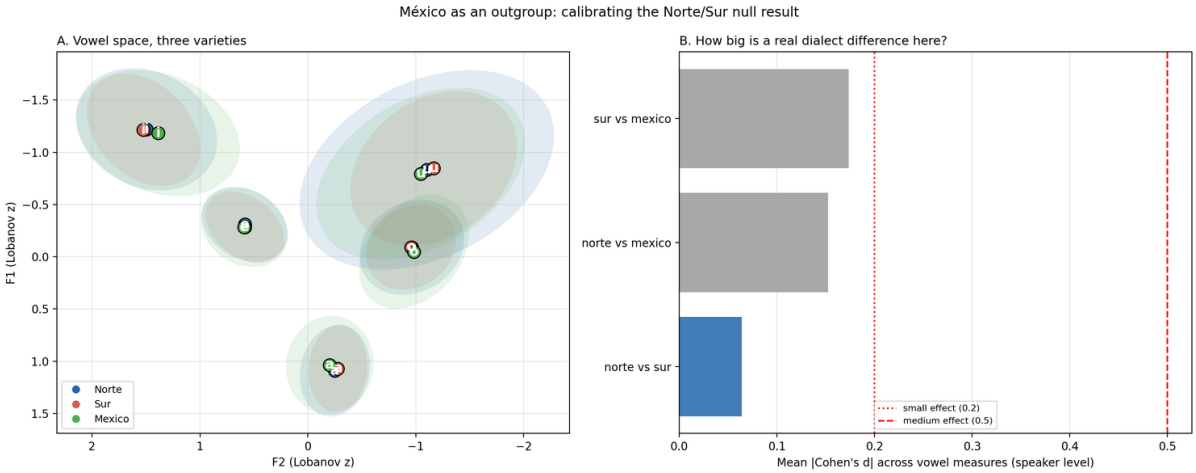



Figure 4: Calibration with an external group. The three varieties overlaid in the normalized vowel space and the average magnitude of the effect for each pairwise contrast. The peninsular contrast is consistently the smallest of the three.

| Position | Measurement | North | South | $d$ | $p$ (Holm) |
|---|---|---|---|---|---|
| Coda | CoG (Hz) | 3,118.7 | 2,417.1 | $+0.56$ | $<0.001$ |
| Coda | Duration | – | – | $+0.30$ | 0.019 |
| Coda | RMS | – | – | $+0.07$ | 1.000 |
| Onset | CoG (Hz) | 3,517.0 | 3,450.1 | $+0.06$ | 1.000 |
| Onset | Duration | – | – | $-0.27$ | 0.033 |
| Onset | RMS | – | – | $+0.03$ | 1.000 |

Table 5: /s/ in North and South, per syllabic position, in speaker level. The dialectal contrast appears in the coda and disappears in the onset.

890 speakers, of which 12,269 are coda segments and 9,370 are onset segments. For each segment, the spectral center of gravity (CoG) is measured along with duration and root-mean-square energy (Jongman et al., 2000).

The pattern is the one predicted by the hypothesis, not the one that would produce a general difference in sibilant production (Figure 5). In coda, the Southern CoG is 700 Hz lower than the Northern one, with $d = +0.56$. At onset, the difference between the same varieties is 67 Hz, with $d = +0.06$ and $p = 1.000$. The interaction between dialect and position on the CoG is significant ($+196.9$ Hz, $p = 0.006$): the South does not produce distinct sibilants; rather, it produces distinct codas, which is the definition of a phonological process conditioned by position.

The contrast can be summarized in a single measure per speaker, the weakening index, defined as the difference between the CoG of their onsets and that of their codas: the higher the value, the more the coda is weakened relative to the full realization by the same speaker. The three varieties are ranked as follows: North, $+359$ Hz over 369 speakers; Mexico, $+836$ Hz over 249; and South, $+1,022$ Hz over 165.

When comparing the South versus the North and Mexico combined, $d = +0.63$ with $p < 0.0001$: the strongest effect in the entire study, and the only finding that dialectology would have predicted in advance. The ranking also replicates what has been described in the literature without having been imposed by the study design: Northern peninsular Spanish is the most conservative variety in coda position, Southern peninsular Spanish is the most innovative, and Mexican Spanish occupies an intermediate position, consistent with a distribution in which coastal varieties aspirate and highland varieties maintain (Hualde, 2005; Arias Álvarez, 2007).

This is the positive result of the study; its methodological reading is taken up in Section 7.

### 6.4 The vowel before coda /s/

The compensatory accounts introduced in Section 2.1 (Alonso et al., 1950) can now be tested directly, by contrasting within each speaker the vowels that precede a coda /s/ with the same vowels in other contexts. A total of 815 speakers provided data in both contexts: 397 from the North, 168 from the South, and 250 from Mexico.

The interaction between dialect and context is significant for F1 ($+0.101$, $p = 0.005$), for F2 ($+0.070$, $p = 0.036$), and for duration ($+0.003$ s, $p = 0.036$). There is, therefore, a vowel effect

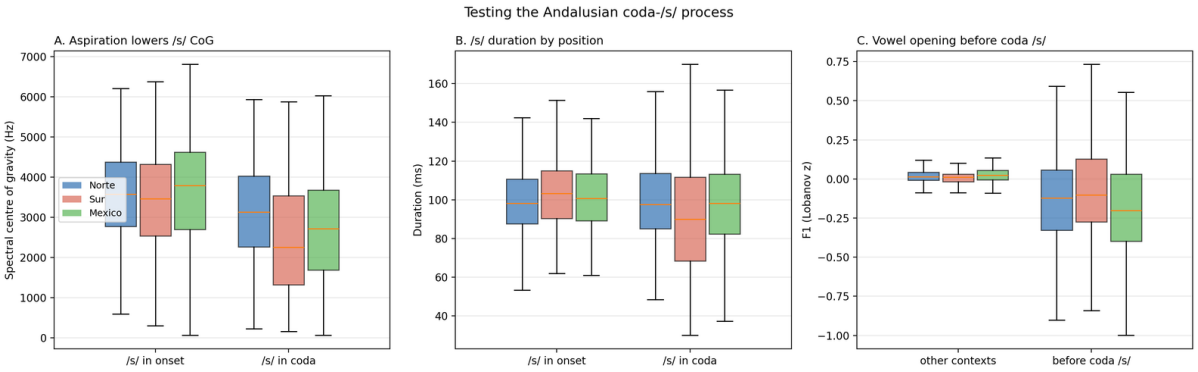



Figure 5: Weakening of /s/ in coda: Spectral center of gravity and duration of /s/ according to syllabic position in the three varieties, and F1 of the preceding vowel according to context. The distinction between varieties appears only in the coda.

associated with the /s/ coda context that differs across varieties.

Its interpretation, however, requires precision, because it does not match what is described in the literature. The intra-speaker shift in F1 when moving to a /s/ coda context is $-0.146$ in the North and $-0.086$ in the South, in Lobanov z-units. Both values are negative: both varieties close the vowel before the /s/ coda, and the South simply closes it less. There is no Southern vowel opening in these data; there is a generalized closing in which the South participates to a lesser extent. The effect is also small, $d = -0.17$, and does not reach significance on its own ($p = 0.063$). Nor is compensatory lengthening observed: the difference in duration goes in the same direction but with a smaller magnitude ($d = -0.10$, $p = 0.241$).

It is worth noting how this result changed with sample size. With 34 Southern speakers, the same contrast yielded $d = -0.83$, a large effect that fit well with the traditional description. With 168 speakers, it is $-0.17$. The interaction remains significant, but the magnitude has been reduced to one-fifth. This is a textbook case of regression to the mean: small samples do not produce biased effects in a specific direction; rather, they produce extreme effects, and those that survive the filter of attention are systematically the largest. It is the same phenomenon that Section 5 documents from another angle, and it is worth keeping this in mind when reading any phonetic study based on just a few dozen speakers.

### 6.5 Exploratory analyses

Two unsupervised analyses round out the picture and, incidentally, serve to correct certain conclusions from the preliminary analysis.

By applying $k$-means as implemented in scikit-learn (Pedregosa et al., 2011) to the normalized formants and evaluating the resulting partition against the actual vowel labels using the adjusted Rand index (Hubert and Arabie, 1985), normalization improves the recovery of the vowel categories (ARI = 0.424 with Lobanov versus 0.332 in raw hertz) while the inclusion of bandwidths degrades it (0.357 and 0.258, respectively). Bandwidths do not contribute to vowel structure, they dilute it.

The preliminary analysis had concluded, based on the silhouette coefficient, that Spanish is organized into "three perceptual zones" rather than five vowels. The $k$-scan reproduces that result, since the silhouette does peak at $k = 3$ (0.454), but the ARI places it at $k = 4$ (0.479), and the actual number of categories is five. The discrepancy has a technical, rather than phonological, explanation: the silhouette measure favors partitions with few compact clusters and decreases mechanically as dimensions are added, so it is not comparable across spaces of different dimensionality; the ARI, being measured against a known truth, is comparable (Rousseeuw, 1987). The $k = 3$ result was a property of the metric, not of Spanish.

When training four classifiers to predict dialect based on acoustic parameters, using speaker-disjoint cross-validation folds, tthe balanced accuracy is 0.500 in all four cases, that is, exactly random (Figure 6). Contrasts that include Mexico are only slightly higher (0.527 and 0.530).

That this reflects the problem and not the characteristics is demonstrated by the positive control: the same models, applied to the same data and using the same partitions, predict the vowel identity

of never-before-seen speakers with an accuracy of 0.762 compared to a chance accuracy of 0.200, that is, 3.8 times chance. The measures contain abundant phonetic information, but what they do not contain is dialectal information.

The experiment also provides a second demonstration of the argument in Section 5. If the partitions are made randomly across tokens, allowing the same speaker to appear in both the training and test sets, the accuracy in dialect prediction is inflated by $+0.20$ on a sample constructed like the one in the preliminary analysis. On the current sample, the same comparison yields an increase of $+0.004$. Speaker identity leakage only distorts the results when a few speakers dominate the sample. When the design is balanced, both validation schemes converge.

## 7 Discussion

In the vowel space, Northern and Southern Peninsular Spanish are indistinguishable at speaker level. In the treatment of coda /s/ they separate clearly, with the onset serving as a null control. The large effects that the preliminary analysis had located in the vowels did not exist at all.

The fact that one of the three results is positive and the other two are null is no coincidence, and the reason lies in the design rather than in dialectology. The contrast that survives is the only one that occurs within the speaker. Comparing the /s/ in coda with the /s/ at the onset of the same speaker instantly neutralizes everything in this corpus that is confounded with dialect: the anatomy of the vocal tracts, the microphone, room noise, age, gender, and reading style. Comparisons between groups, on the other hand, must account for all these factors through normalization, statistical control, and balanced sampling. The practical lesson is that, in a corpus of volunteers, a well-chosen intra-speaker contrast is more valuable than any statistical refinement applied to an overall comparison.

This also delimits what Common Voice is good for in phonetics. The unevenness described in Section 2.3 is not a defect to be tolerated but the single fact that governs how a study on this corpus must be designed. Consequently, the speaker identifier is not merely incidental metadata but the variable that determines whether an analysis is valid. Any study that samples clips without grouping them by speaker runs the risk of measuring individuals while believing it is measuring varieties, and this risk increases as the corpus grows larger, because the volume of data increases statistical significance without increasing independence. With per-speaker caps and prior aggregation, the same corpus supports robust inference.

One caveat remains. Section 6.1 established no difference is detectable, which is not the same as establishing that none exists, and the difference between those two statements matters here. The descriptive literature does not describe the Southern vowel system as slightly different from the Northern one; it describes it as audibly and systematically different. That stronger claim is the one these data fail to support, in the vowel space as a whole and also in the compensatory opening before coda /s/, where the effect is real but five times smaller than a small sample had suggested. The claim they do support is the consonantal one, and it carries the largest effect in the study.

## 8 Limitations and conclusions

The data consists of read speech produced by volunteers under uncontrolled recording conditions; therefore, the conclusions do not automatically apply to spontaneous speech. Dialect labels are self-reported and unverifiable, and their granularity is coarse: the label “Southern Peninsula” encompasses Andalusian, Extremaduran, and Murcian varieties whose phonetic behavior is not homogeneous, which necessarily attenuates any contrast. The sex imbalance is mitigated through normalization but does not disappear.

In terms of measurement, validation with synthetic vowels reveals a bias of +33.7% in the F1 estimated for /u/, so no conclusions should be based on that vowel; the bias is identical in both dialects and cannot generate spurious differences, but it can introduce noise. In the analysis of /s/, it is important to bear in mind that forced alignment sets a boundary for the segment even if it has been completely elided: a total elision is therefore measured as a segment with a very low center of gravity rather than as an absence, which likely underestimates rather than exaggerates the Southern weakening. Finally, the null result for vowel space constrains but does not eliminate: differences below $d \approx 0.24$ remain compatible with these data.

Three conclusions, in the order the study reached them.

First, the large effects in the initial analysis (F1 at $d = 1.34$ and B2 at $d = -1.43$) came from the

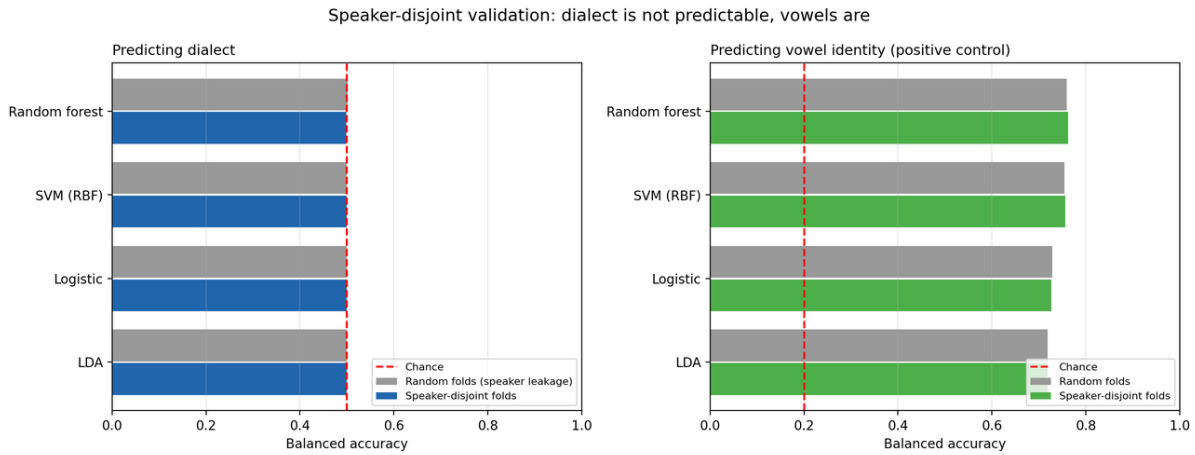



Figure 6: Dialect prediction against vowel identity prediction with speaker-disjoint cross-validation and with random partitions. Dialect sits at chance under both schemes in the balanced sample. Vowel identity is recovered far above chance for previously unseen speakers.

sampling rather than from the dialects. Two speakers supplied 54% and 80% of their respective cells, and the size of the effects tracks how much those two are allowed to contribute. Once the speaker becomes the unit of inference, nothing is left of them.

Second, across 461 Northern and 186 Southern speakers I find no detectable difference in the vowel space, in a design that would have caught anything at $d \geq 0.24$. The Mexican outgroup rules out the obvious alternative explanation: the same procedure, applied to a contrast nobody disputes, produces effects 2.5 times larger.

Third, what separates these two varieties is consonantal. Southern speakers weaken coda /s/ substantially and consistently ($d = +0.63$), onset /s/ shows nothing, and the three varieties line up as the dialectological literature would predict. This was also the only contrast I tested within speakers rather than between groups, and I doubt that is a coincidence. On a corpus like this one, a comparison that each speaker controls for themselves is worth more than a more elaborate analysis of one that nobody controls.

## References


Patti Adank, Roel Smits, and Roeland van Hout. 2004. A comparison of vowel normalization procedures for language variation research. *The Journal of the Acoustical Society of America*, 116(5):3099–3107.

Dámaso Alonso, Alonso Zamora Vicente, and María Josefa Canellada de Zamora. 1950. Vocales andaluzas. contribución al estudio de la fonología peninsular. *Nueva Revista de Filología Hispánica*, 4(3):209–230.

Manuel Alvar. 1996. *Manual de dialectología hispánica: el español de España*. Ariel, Barcelona.

Rosana Ardila, Megan Branson, Kelly Davis, et al. 2020. Common Voice: A massively-multilingual speech corpus. In *Proceedings of the Twelfth Language Resources and Evaluation Conference*, pages 4218–4222, Marseille, France. European Language Resources Association.

Beatriz Arias Álvarez. 2007. Estudio sobre la realización de algunas consonantes en coda en el habla popular de la Ciudad de México: hacia una interpretación causal de los hechos. *Boletín de Filología*, 42:11–35.

Dale J. Barr, Roger Levy, Christoph Scheepers, and Harry J. Tily. 2013. Random effects structure for confirmatory hypothesis testing: Keep it maximal. *Journal of Memory and Language*, 68(3):255–278.

Paul Boersma and David Weenink. 2021. Praat: doing phonetics by computer (version 6.1.38) [computer program]. Version 6.1.38, 2 January 2021, as bundled with Parselmouth 0.4.7.

Jacob Cohen. 1988. *Statistical Power Analysis for the Behavioral Sciences*, 2 edition. Lawrence Erlbaum Associates, Hillsdale, NJ.

Bradley Efron and Robert J. Tibshirani. 1993. *An Introduction to the Bootstrap*. Chapman and Hall, New York.

Chip Gerfen. 2002. Andalusian codas. *Probus*, 14(2):247–277.

Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, et al. 2020. Array programming with NumPy. *Nature*, 585(7825):357–362.

Sture Holm. 1979. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70.

José Ignacio Hualde. 2005. *The Sounds of Spanish*. Cambridge University Press, Cambridge.

Lawrence Hubert and Phipps Arabie. 1985. Comparing partitions. *Journal of Classification*, 2(1):193–218.

John D. Hunter. 2007. Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, 9(3):90–95.

Stuart H. Hurlbert. 1984. Pseudoreplication and the design of ecological field experiments. *Ecological Monographs*, 54(2):187–211.

Yannick Jadoul, Bill Thompson, and Bart de Boer. 2018. Introducing Parselmouth: A Python interface to Praat. *Journal of Phonetics*, 71:1–15.

Allard Jongman, Ratree Wayland, and Serena Wong. 2000. Acoustic characteristics of English fricatives. *The Journal of the Acoustical Society of America*, 108(3):1252–1263.

B. M. Lobanov. 1971. Classification of Russian vowels spoken by different speakers. *The Journal of the Acoustical Society of America*, 49(2B):606–608.

Eugenio Martínez Celdrán and Ana María Fernández Planas. 2007. *Manual de fonética española: articulaciones y sonidos del español*. Ariel, Barcelona.

Michael McAuliffe, Michaela Socolof, Sarah Mihuc, Michael Wagner, and Morgan Sonderegger. 2017. Montreal forced aligner: Trainable text-speech alignment using Kaldi. In *Interspeech 2017*, pages 498–502. ISCA.

Wes McKinney. 2010. Data structures for statistical computing in Python. In *Proceedings of the 9th Python in Science Conference*, pages 56–61, Austin, Texas.

Antonio Narbona Jiménez, Rafael Cano Aguilar, and Ramón Morillo-Velarde Pérez. 1998. *El español hablado en Andalucía*, 1 edition. Ariel lingüística. Ariel, Barcelona.

Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, et al. 2011. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830.

Daniel Povey, Arnab Ghoshal, Gilles Boulianne, et al. 2011. The Kaldi speech recognition toolkit. In *Proceedings of ASRU 2011*, pages 1–4, Hilton Waikoloa Village Resort, Hawaii. IEEE Signal Processing Society.

Antonio Quilis. 1981. *Fonética acústica de la lengua española*. Number 49 in Biblioteca románica hispánica. III. Manuales. Gredos, Madrid.

Peter J. Rousseeuw. 1987. Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20:53–65.

Hanna Ruch and Jonathan Harrington. 2014. Synchronic and diachronic factors in the change from pre-aspiration to post-aspiration in Andalusian Spanish. *Journal of Phonetics*, 45:12–25.

Skipper Seabold and Josef Perktold. 2010. Statsmodels: Econometric and statistical modeling with Python. In *Proceedings of the 9th Python in Science Conference*, pages 92–96.

Francisco Torreira. 2007. Coarticulation between aspirated-s and voiceless stops in Spanish: An interdialectal comparison. In *Selected Proceedings of the 9th Hispanic Linguistics Symposium*, pages 113–120, Somerville, MA. Cascadilla Press.

Hartmut Traunmüller. 1990. Analytical expressions for the tonotopic sensory scale. *The Journal of the Acoustical Society of America*, 88(1):97–100.

Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, et al. 2020. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods*, 17:261–272.

Bodo Winter. 2019. *Statistics for Linguists: An Introduction Using R*, 1 edition. Routledge, New York.